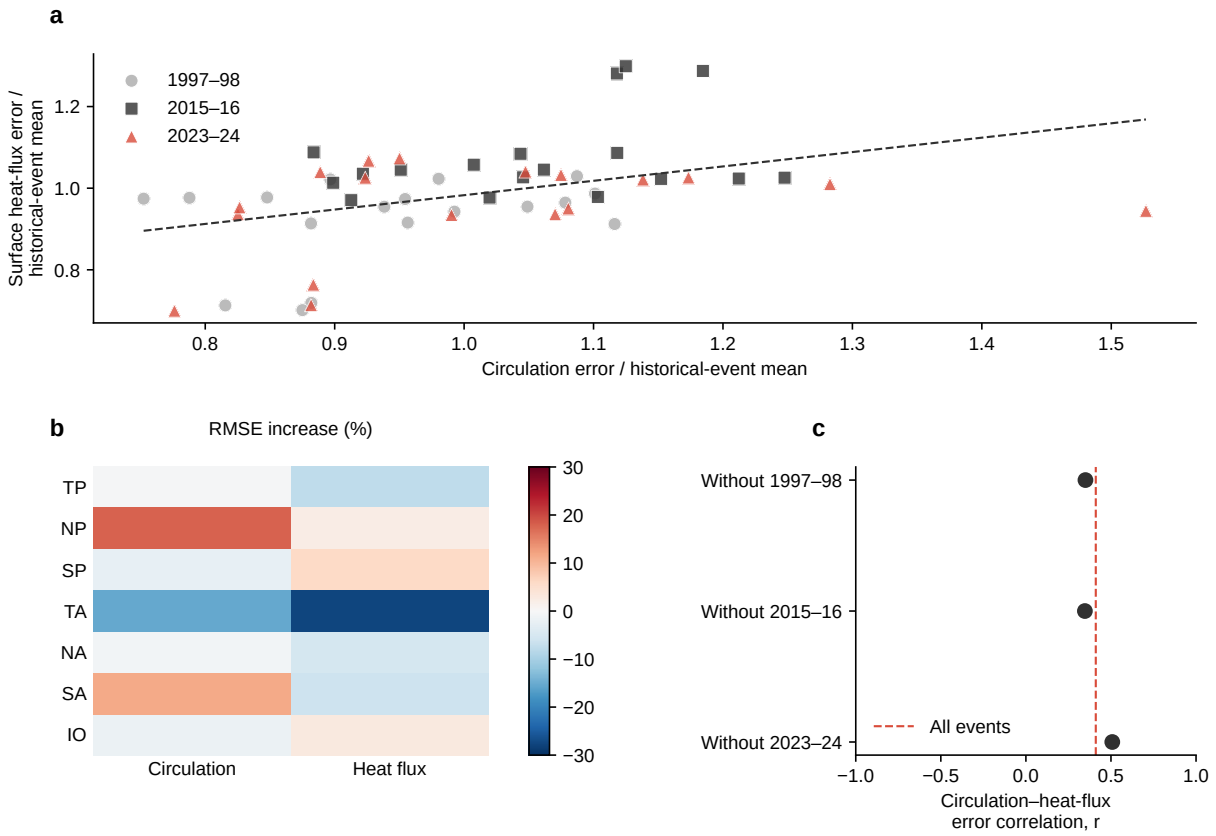
Supplementary Fig. 14 | CFSv2-only circulation and surface heat-flux error comparison



a, CFSv2 circulation RMSE versus net surface heat-flux RMSE, each divided by its own 1997–1998/2015–2016 mean at the same region and lead. Colours and symbols distinguish the three events. The dashed line is the pooled fit. b, Percentage RMSE increases in 2023–2024 relative to those earlier-event means, averaged over leads 1, 3 and 6 months. Red denotes greater error and blue smaller error. c, Correlations between the two normalized errors after omitting each event; the red dashed line denotes the pooled estimate. All masks now use the ±23.5° divisions of main Figs. 3–4. The 54 remote pairs comprise three events × six recipient regions × three leads; nine additional tropical-Pacific pairs enter a.

This is a separately downloaded and processed CFSv2-only check, not an independent model: CFSv2 also contributes to the main SST ensemble. Circulation is departure from contemporaneous zonal mean, verified against ERA5; net heat flux is verified against ORAS5 and is positive into the ocean. Four members and three DJF months are equally weighted. The circulation metric differs from the anomaly correlation in main Fig. 3; no causal or independent-event significance is inferred.